ENVS-193DS Final

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Link to Git Hub

GitHub Repository

Set Up

```
library(tidyverse)   # data wrangling and ggplot
library(here)        # file paths
library(janitor)     # data cleaning
library(DHARMa)      # diagnostic checks for models
library(MuMIn)       # AIC model selection
library(ggeffects)   # model predictions
library(flextable)   # making tables
```

```
nest_data <- read_csv(here("data", "nest_data_final.csv"))
```

```
Rows: 299 Columns: 10
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (3): alive, bird_species, ant_species
dbl (7): height_cm, diameter_mm, case_control, case, canopy_width, canopy_le...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Problem 1. Research writing

1a. Transparent statistical methods

In part 1, they used a logistic regression, since the response variable (probability of American Avocet micro habitat use) is binary and distance to water edge is a continuous predictor. In part 2, they used a chi-square test of independence, since they are testing the relationship between two categorical variables (bird species and habitat type).

1b. Suggestions for rewriting

Part 1: Distance to water edge was a significant predictor of American Avocet microhabitat use, meaning that where a bird was found was related to how far it was from the water's edge.

Part 2: The three bird species — American Avocet, Forster's Tern, and Black-necked Stilt — were not evenly distributed across habitat types (managed pond, non-tidal marsh, salt pond, tidal marsh, and tidal mudflat), suggesting that each species shows preferences for particular habitats.

1c. Further analysis

Presence of wet bare ground (categorical: present or absent, assessed visually during point count surveys) could influence the probability of American Avocet microhabitat use because, according to Schacter et al. (2023), Avocets fed primarily on wet bare ground in 65% of foraging observations, suggesting they strongly select for this substrate type when choosing where to forage. Sites lacking wet bare ground would therefore be expected to have lower probability of Avocet microhabitat use.

Problem 2. Data analysis

2a. Clean and Wrangle

```
nests_clean <- nest_data |>
  # select only the 3 columns of interest
  select(case_control, height_cm, ant_species) |>
  # filter to only the 3 ant species of interest
  filter(ant_species %in% c("AB", "RRB", "BBR")) |>
  # create column with full scientific names
  mutate(ant_species_full = case_when(
    ant_species == "AB"  ~ "Crematogaster sjostedti",
```

```r
    ant_species == "RRB" ~ "Crematogaster mimosae",
    ant_species == "BBR" ~ "Crematogaster nigriceps"
  )) |>
  # reorder factor levels: C. sjostedti, C. mimosae, C. nigriceps
  mutate(ant_species_full = fct_relevel(ant_species_full,
                                        "Crematogaster sjostedti",
                                        "Crematogaster mimosae",
                                        "Crematogaster nigriceps"))

# display structure of nests_clean
str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ case_control    : num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ height_cm       : num [1:270] 392 310 513 154 324 ...
 $ ant_species     : chr [1:270] "RRB" "RRB" "RRB" "RRB" ...
 $ ant_species_full: Factor w/ 3 levels "Crematogaster sjostedti",..: 2 2 2 2 1 2 3 2 1 3 ..
```

```r
# display 10 random rows
slice_sample(nests_clean, n = 10)
```

```
# A tibble: 10 x 4
   case_control height_cm ant_species ant_species_full
          <dbl>     <dbl> <chr>       <fct>
 1            0      154. RRB         Crematogaster mimosae
 2            0      103  RRB         Crematogaster mimosae
 3            0      238. RRB         Crematogaster mimosae
 4            0      286  AB          Crematogaster sjostedti
 5            0      112  BBR         Crematogaster nigriceps
 6            0      212. RRB         Crematogaster mimosae
 7            0      104  RRB         Crematogaster mimosae
 8            0      105  BBR         Crematogaster nigriceps
 9            0      132  RRB         Crematogaster mimosae
10            0      118. RRB         Crematogaster mimosae
```

2b. Response variable

In this dataset, a 1 indicates that the tree was a "used" tree, meaning a bird nest of any species was found in that tree. A 0 indicates that the tree was an "available" tree, meaning it was one of the four nearest neighbors to a nest tree but did not itself contain a nest.

2c. Purpose of study

Tree height is a continuous variable measured in centimeters. As tree height increases, the probability of nest occurrence would increase because taller trees may offer greater structural complexity and concealment from predators, making them more suitable nest sites; the authors found that for each 1 m increase in height, the odds that birds nested in a tree increased by 20%. This information was found in the Methods section (paragraph 2) and the Results section (paragraph 1).

Acacia ant species is a categorical variable with three levels: Crematogaster sjostedti, Crematogaster mimosae, and Crematogaster nigriceps. As ant species identity shifts from less aggressive defenders (C. sjostedti) to more aggressive defenders (C. mimosae and C. nigriceps), the probability of nest occurrence would increase because more aggressive ant species vigorously defend host trees against mammalian herbivores and predators, which may also protect bird nests from disturbance and predation. This information was found in the Introduction (paragraphs 2–3) and the Discussion (paragraphs 1–2).

2d. Table of Models

```r
model_table <- data.frame(
  "Model Number" = c("Model 1", "Model 2"),
  "Response Variable" = c("Probability of nest occurrence",
                          "Probability of nest occurrence"),
  "Predictor Variables" = c("Tree height (cm), ant species",
                            "Tree height (cm), ant species"),
  "Interaction Term" = c("No", "Yes (height × ant species)")
)

flextable(model_table) |>
  set_header_labels(
    Model.Number = "Model Number",
    Response.Variable = "Response Variable",
    Predictor.Variables = "Predictor Variables",
    Interaction.Term = "Interaction Term"
  ) |>
  align(align = "center", part = "all") |>
  bold(part = "header") |>
  set_table_properties(width = 1, layout = "autofit") |>
  fontsize(size = 9, part = "all")
```

Model Number	Response Variable	Predictor Variables	Interaction Term
Model 1	Probability of nest occurrence	Tree height (cm), ant species	No
Model 2	Probability of nest occurrence	Tree height (cm), ant species	Yes (height × ant species)

2e. Run models

```
# model 1: tree height + ant species, no interaction
model1 <- glm(case_control ~ height_cm + ant_species_full,
              data = nests_clean,
              family = binomial)

# model 2: tree height * ant species, with interaction
model2 <- glm(case_control ~ height_cm * ant_species_full,
              data = nests_clean,
              family = binomial)
```

2f. Select the best model

The best model as determined by Akaike's Information Criterion (AIC) was the model with an interaction term between tree height and acacia ant species, predicting the probability of bird nest occurrence (AICc = 238.12). This model performed better than the additive model with no interaction term (AICc = 258.89), indicating that the effect of tree height on the probability of nest occurrence differs depending on which acacia ant species occupies the tree.